If you add further alternating layers of interactivity but with graphical placement above and below, then you can end up making a `<canvas>` and `<svg>` layer cake in your DOM that's hard to manage and also hard to mentally conceptualize.

11.2 Big network data

It's great that `d3.geo.path` has built-in functionality for drawing geodata to canvas, but what about other types of data visualization? One of the most performance-intensive layouts is the force-directed layout that we dealt with in chapter 6. The layout calculates new positions for each node in your network at every tick. When you use SVG, you need to redraw the network constantly. When I first started working with force-directed layouts in D3, I found that any network with more than 100 nodes was too slow to prove useful. That was a problem because larger networks could still have structure that would benefit from interactivity and animation that needed SVG.

In my own work, I looked at how different small D3 applications hosted on gist.github.com share common D3 functions. D3 coders can understand how different information visualization methods use D3 functions commonly associated with other types of information visualization. You can explore this network along with how D3 Meetup users describe themselves at http://emeeks.github.io/introspect/block_block.html.

To explore these connections, I needed to have a method for dealing with over a thousand different examples and thousands of connections between them. You can see some of this network in figure 11.8. I wanted to show how this network changed

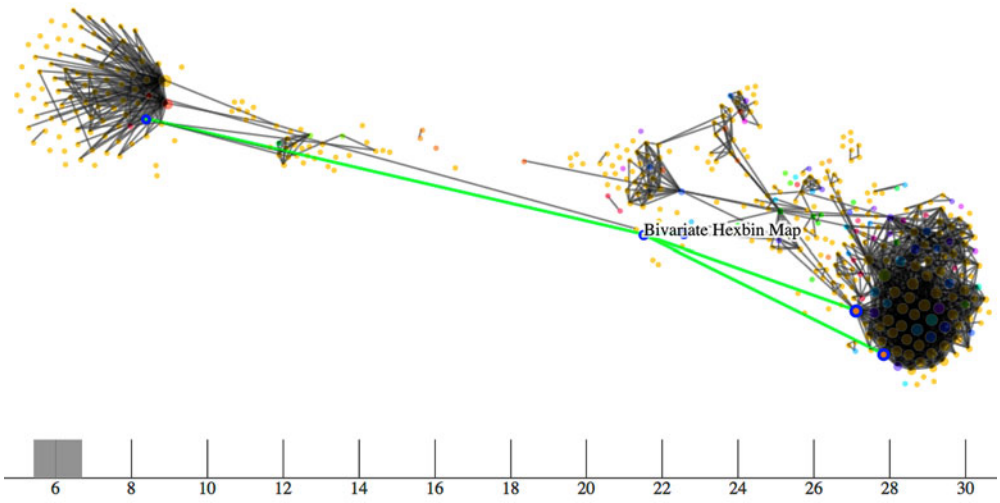


Figure 11.8 A network of D3 examples hosted on gist.github.com that connects different examples to each other by shared functions. Here you can see that the example "Bivariate Hexbin Map" by Mike Bostock (http://bl.ocks.org/mbostock/4330486) shares functions in common with three different examples: "Metropolitan Unemployment," "Marey's Trains II," and "GitHub Users Worldwide." The brush and axis component allows you to filter the network by the number of connections from one block to another.